

Basic Asymptotic Theory

This handout summarizes some definitions and limit theorems which you can find in Chapter 3 of Wooldridge (2012) and in Chapter 2 of Hayashi (2000) that are important for studying the proofs of MLE.

1 Convergence of Deterministic Sequences

Definition 1.

1. A sequence of nonrandom numbers $\{x_n : n = 1, 2, \dots\}$ converges to x (has a limit x) if $\forall \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}$ such that if $n > N_\varepsilon$, then $|x_n - x| < \varepsilon$. We write $x_n \rightarrow x$ as $n \rightarrow \infty$.
2. A sequence of nonrandom numbers $\{x_n : n = 1, 2, \dots\}$ is bounded if and only if there is some $b < \infty$ such that $|x_n| \leq b$ for all $n = 1, 2, \dots$.

Definition 2.

1. A sequence $\{x_n\}$ is $O(N^\lambda)$ (at least of order N^λ) if $N^{-\lambda}x_n$ is bounded. When $\lambda = 0$, $\{x_n\}$ is bounded, and we also write $x_n = O(1)$ (big oh one).
2. $\{x_n\}$ is $o(N^\lambda)$ if $N^{-\lambda}x_n \rightarrow 0$. When $\lambda = 0$, $\{x_n\}$ converges to zero, and we also write $x_n = o(1)$ (little oh one).

2 Convergence and Boundedness in Probability

Definition 3.

1. A sequence of random variables $\{x_n : n = 1, 2, \dots\}$ converges in probability to the constant x if for all $\varepsilon > 0$, $P(|x_n - x| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$. We write $x_n \xrightarrow{p} x$ and say x is the probability limit (plim) of x_n : $\text{plim } x_n = x$. In a special case where $x = 0$, we also say $\{x_n\}$ is $o_p(1)$ (little oh p one). We also write $x_n = o_p(1)$ or $x_n \xrightarrow{p} 0$.

2. A sequence of random variables $\{x_n\}$ is bounded in probability if and only if for every $\varepsilon > 0$, there exists a $b_\varepsilon < \infty$ and an integer N_ε such that $P(|x_n| \geq b_\varepsilon) < \varepsilon$ for all $N \geq N_\varepsilon$. We write $x_n = O_p(s)$ ($\{x_n\}$ is big oh p one).

Lemma 1. If $x_n \xrightarrow{p} x$, then $x_n = O_p(1)$, i.e., $\{x_n\}$ is bounded in probability.

Lemma 2. If $w_n = o_p(1)$, $x_n = o_p(1)$, $y_n = O_p(1)$, and $z_n = O_p(1)$, then

1. $w_n + x_n = o_p(1)$.
2. $y_n + z_n = O_p(1)$.
3. $y_n z_n = O_p(1)$.
4. $x_n z_n = o_p(1)$.

Lemma 3. Let $\{\mathbf{Z}_n : n = 1, 2, \dots\}$ be a sequence of $J \times K$ matrices such that $\mathbf{Z}_n = o_p(1)$, and let $\{x_n\}$ be a sequence of $J \times 1$ random vectors such that $x_n = O_p(1)$. Then $\mathbf{Z}'_n x_n = o_p(1)$.

Lemma 4 (Slutsky's theorem). Let $g : \mathbb{R}^K \rightarrow \mathbb{R}^J$ be a function continuous at some point $c \in \mathbb{R}^K$. Let $\{x_n : n = 1, 2, \dots\}$ be a sequence of $K \times 1$ random vectors such that $x_n \xrightarrow{p} c$. Then, $g(x_n) \xrightarrow{p} g(c)$ as $n \rightarrow \infty$. In other words, $\text{plim } g(x_n) \rightarrow g(\text{plim } x_n)$.

3 Convergence in Distribution

Definition 4. A sequence of random variables $\{x_n : n = 1, 2, \dots\}$ converges in distribution to the continuous random variable x if and only if $F_{x_n}(\xi) \rightarrow F_x(\xi)$ as $n \rightarrow \infty$ for all $\xi \in \mathbb{R}$, where F_{x_n} is the cumulative distribution function (c.d.f) of x_n and F_x is the c.d.f. of x . We write $x_n \xrightarrow{d} x$.

Lemma 5. If $x_n \xrightarrow{d} x$, $y_n \xrightarrow{p} y$, $z_n \xrightarrow{p} 0$, and $A_n \xrightarrow{p} A$, then

1. $x_n + y_n \xrightarrow{d} x + y$.
2. $x_n z_n \xrightarrow{d} 0$.
3. $A_n x_n \xrightarrow{d} Ax$ provided A_n and x_n are conformable. In particular, if $x \sim N(\mathbf{0}, \Sigma)$, then $A_n x_n \xrightarrow{d} N(\mathbf{0}, A\Sigma A')$.
4. $x'_n A_n^{-1} x_n \xrightarrow{d} x' A^{-1} x$ provided A_n and x_n are conformable and A is non-singular.

Lemma 6 (Continuous Mapping Theorem). Let $\{x_n\}$ be a sequence of $K \times 1$ random vectors such that $x_n \xrightarrow{d} x$. If $g : \mathbb{R}^K \rightarrow \mathbb{R}^J$ is a continuous function, then $g(x_n) \xrightarrow{d} g(x)$ as $n \rightarrow \infty$.

4 Limit Theorems

Theorem 1 (Weak Law of Large Numbers). Let $\{x_n : n = 1, 2, \dots\}$ be a sequence of independent, identically distributed $G \times 1$ random vectors such that $E(|x_{ig}| < \infty)$, $g = 1, 2, \dots, G$. Then the sequence satisfies the **weak law of large numbers (WLLN)**:

$$\frac{1}{N} \sum_{n=1}^N x_n \xrightarrow{p} E(x_n).$$

Theorem 2 (Lindeberg-Levy CLT). Let $\{x_n : n = 1, 2, \dots\}$ be a sequence of independent, identically distributed $G \times 1$ random vectors such that $E(|x_{ig}| < \infty)$, $g = 1, 2, \dots, G$. Then the sequence satisfies the **the central limit theorem (CLT)**; that is,

$$\frac{1}{N^{1/2}} \sum_{n=1}^N x_n \xrightarrow{d} \mathbf{Z}$$

where $\mathbf{Z} \sim N(\mathbf{0}, \mathbf{V})$ and $\mathbf{V} = E(x_n x_n')$.

5 Delta Method

Lemma 7 (Delta Method). Suppose $\{\theta_n\}$ is a sequence of K -dimensional random vectors such that $\theta_n \xrightarrow{p} \theta$ and $\sqrt{n}(\theta_n - \theta) \xrightarrow{d} z$, and suppose $g : \mathbb{R}^K \rightarrow \mathbb{R}^J$ has continuous first derivatives with $G(\theta)$ denoting a $J \times K$ matrix of first derivatives evaluated at θ :

$$G(\theta) \equiv \frac{\partial g(\theta)}{\partial \theta'}.$$

Then,

$$\sqrt{n}(g(\theta_n) - g(\theta)) \xrightarrow{d} G(\theta)z.$$

Remarks 1. Note that, in the lecture notes (p.24 of OLS or p.8 of MLE), Dr. Hwang defines $G(\theta)$ as a $K \times J$ matrix of first derivatives evaluated at θ :

$$G(\theta) \equiv \frac{\partial g(\theta)}{\partial \theta}.$$

Thus, the necessary part of the lemma will become

$$\sqrt{n}(g(\theta_n) - g(\theta)) \xrightarrow{d} G(\theta)'z.$$